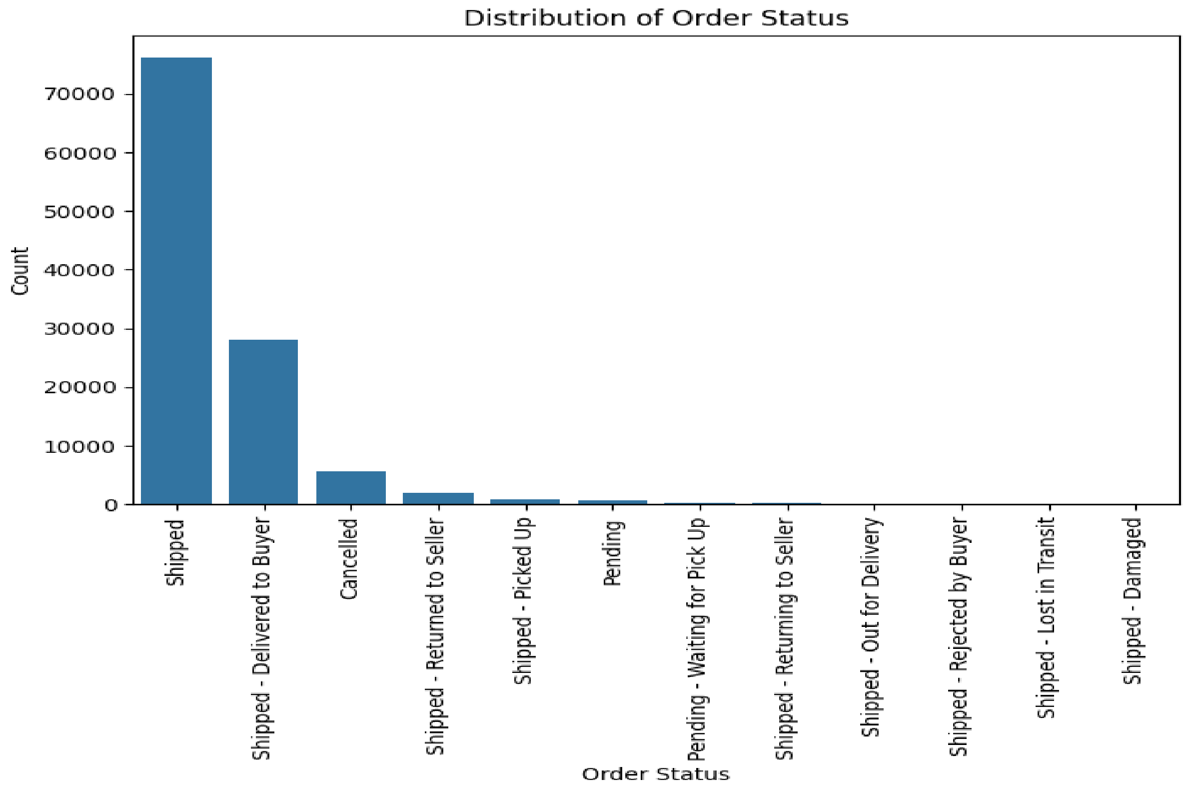
#Task – 29 June 2026:- Perform EDA on the Sales Data ,Perform 5 Viualizations, Write down the 5 insights for these findings.

Visualization 1: Order Status Distribution

```
plt.figure(figsize=(8,5))
sns.countplot(data=df, x='Status', order=df['Status'].value_counts().index)
plt.xticks(rotation=90)
plt.title("Distribution of Order Status")
plt.xlabel("Order Status")
plt.ylabel("Count")
plt.show()
```

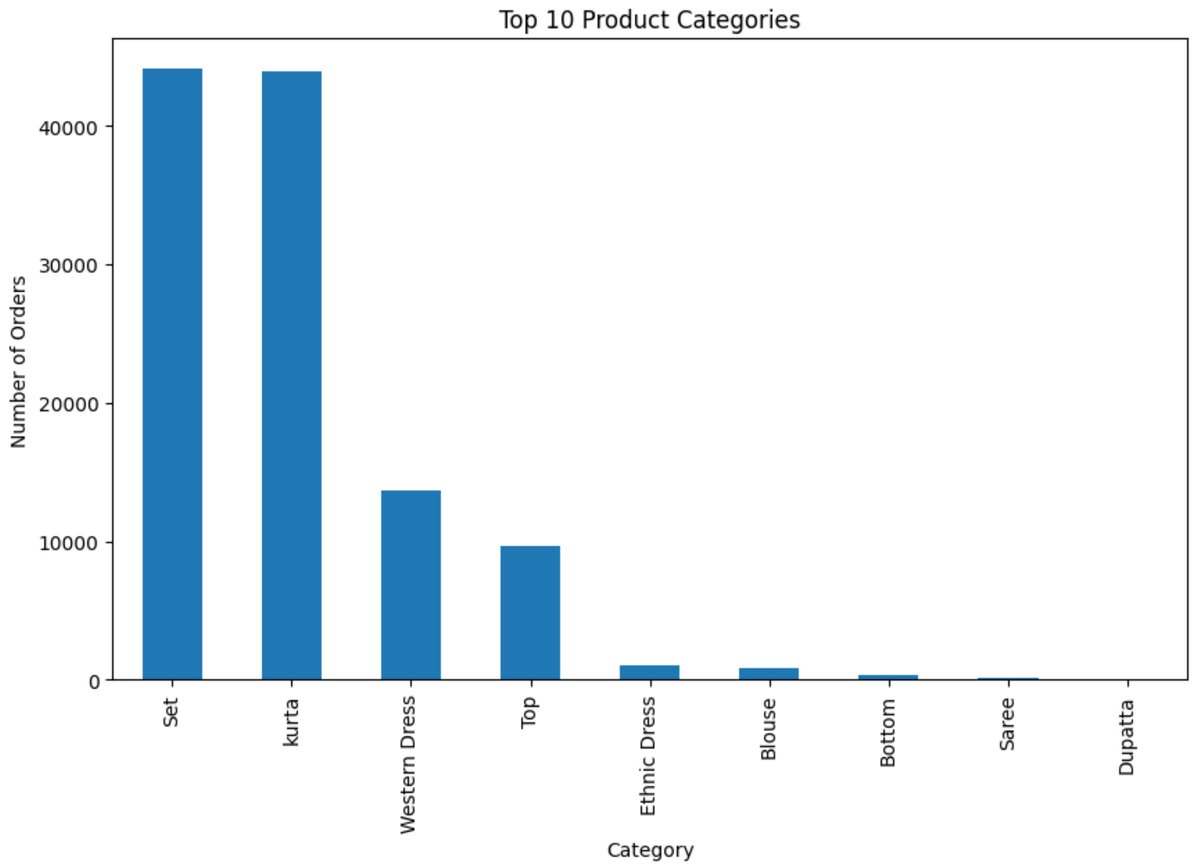


Insight

- Most orders are Shipped, indicating a high order fulfillment rate.
- Cancelled and Returned orders are much lower than shipped orders.

Visualization 2: Top 10 Product Categories

```
plt.figure(figsize=(10,6))
df['Category'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Product Categories")
plt.xlabel("Category")
plt.ylabel("Number of Orders")
plt.show()
```

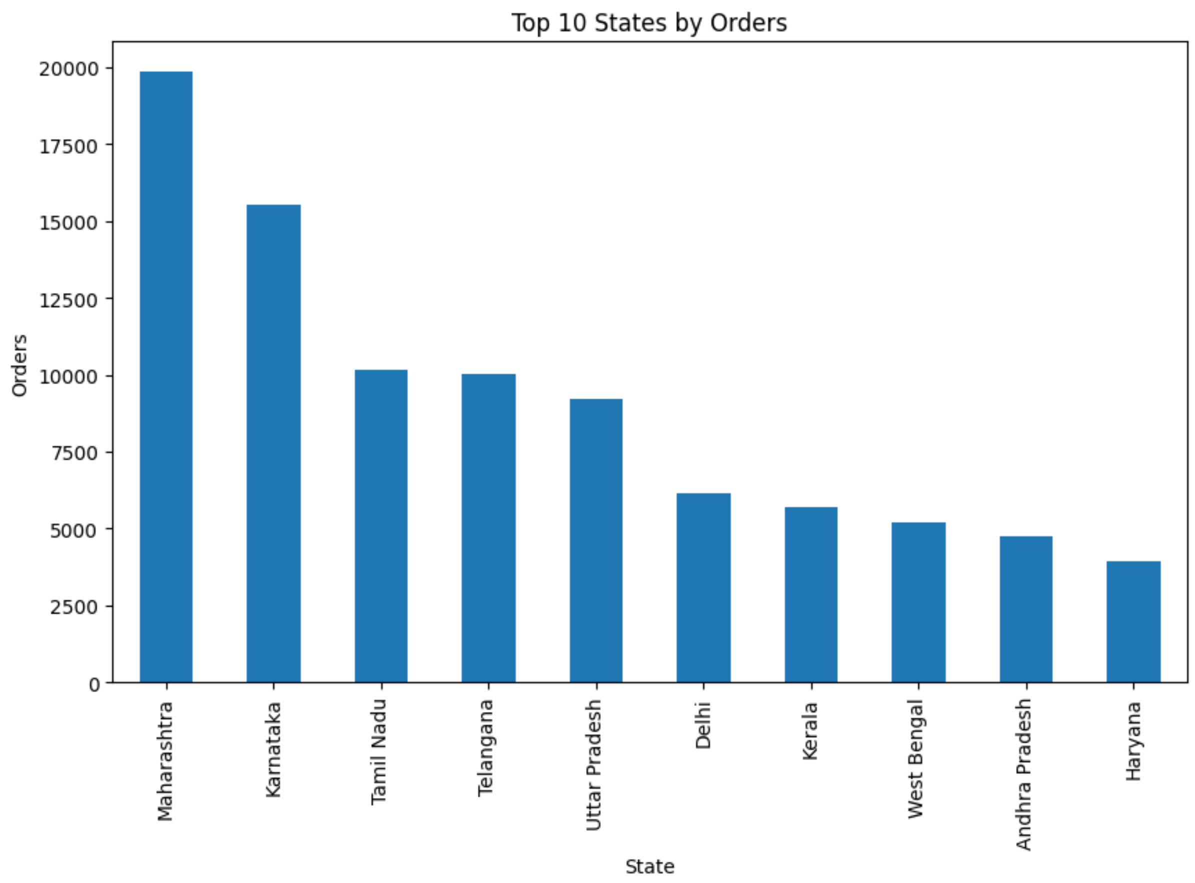


Insight

- Some product categories receive significantly more orders than others.
- These top-selling categories contribute most to overall sales.

Visualization 3: Top 10 States by Orders

```
plt.figure(figsize=(10,6))
df['ship-state'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 States by Orders")
plt.xlabel("State")
plt.ylabel("Orders")
plt.show()
```

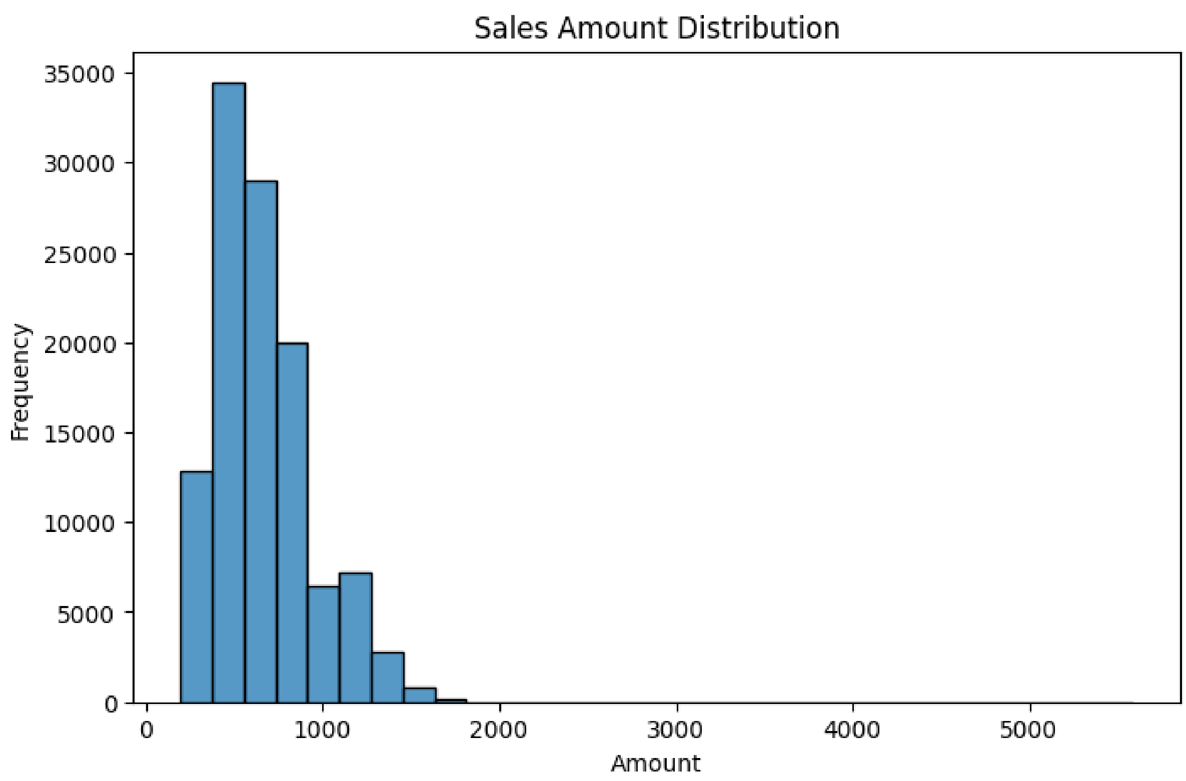


Insight

- A few states generate the majority of orders.
- These regions can be targeted for marketing and inventory planning.

Visualization 4: Sales Amount Distribution

```
plt.figure(figsize=(8,5))
sns.histplot(df['Amount'], bins=30)
plt.title("Sales Amount Distribution")
plt.xlabel("Amount")
plt.ylabel("Frequency")
plt.show()
```

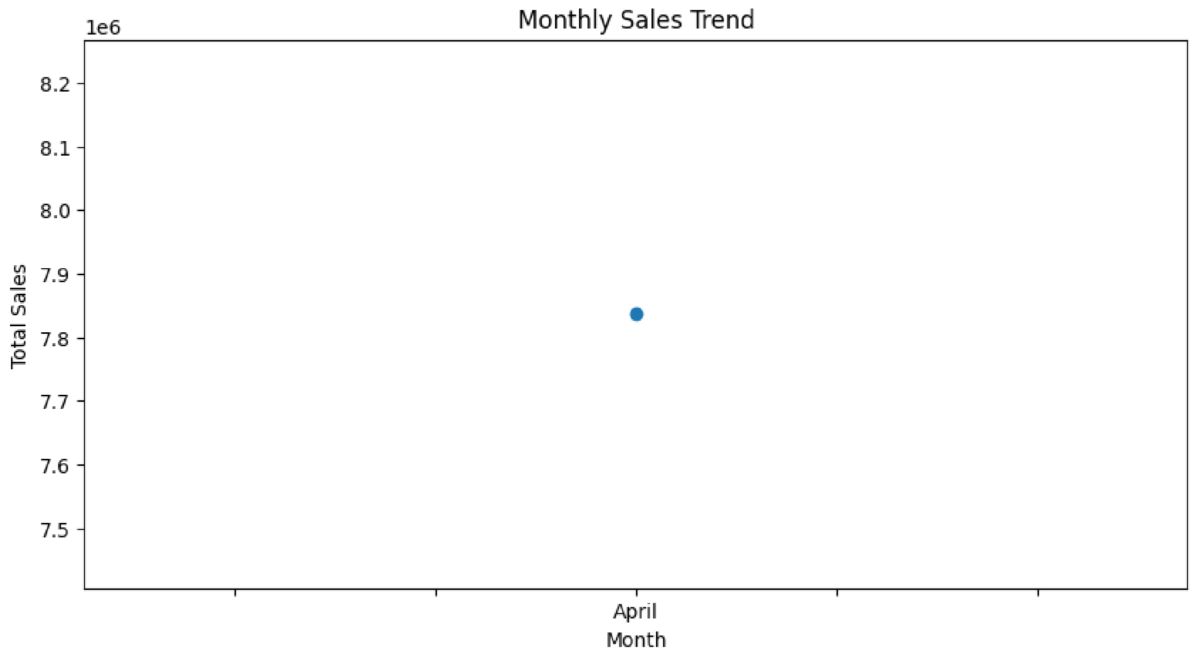


Insight

- Most orders fall within the lower price range.
- Only a small number of high-value orders exist.

Visualization 5: Monthly Sales Trend

```
df['Date'] = pd.to_datetime(df['Date'])
df['Month'] = df['Date'].dt.month_name()
monthly_sales = df.groupby('Month')['Amount'].sum()
plt.figure(figsize=(10,5))
monthly_sales.plot(marker='o')
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.show()
```

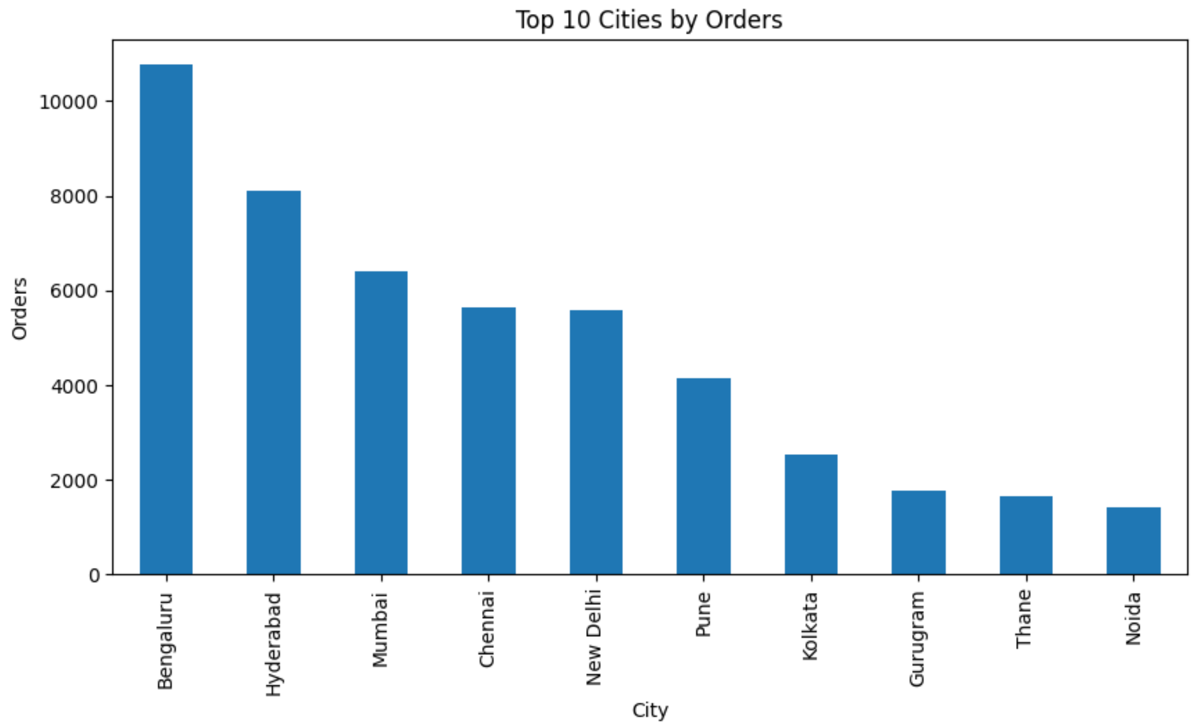


Insight

- Sales fluctuate across months.
- Peak sales months indicate seasonal demand.
- Low sales months may require promotional campaigns.

Visualization 6: Top 10 Cities by order

```
plt.figure(figsize=(10,5))
df['ship-city'].value_counts().head(10).plot(kind='bar')
plt.title("Top 10 Cities by Orders")
plt.xlabel("City")
plt.ylabel("Orders")
plt.show()
```



Insight

- A few cities contribute a large share of orders.
- Businesses can focus logistics and marketing in these cities.